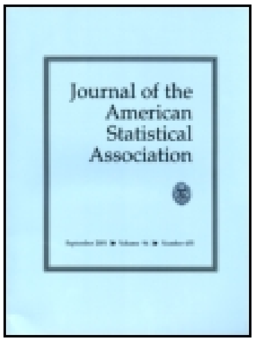

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Sample Size Reestimation for Clinical Trials With Censored Survival Data

Yu SHEN and Jianwen CAI

A flexible design with updating of sample size in clinical trials is proposed for censored survival data. Patients enter trials serially and are subject to random loss to follow-up. The statistical inference is based on a weighted average of the linear rank statistics, where the weight function at each look depends on prior observed data. A stopping rule is devised to allow early termination and acceptance of the null hypothesis when the experimental treatment offers no advantage or is inferior to the control. The null hypothesis that two survival distributions are equal may be rejected only at the last step when the weight function is used up. The overall type I error rate is preserved. Independent increments for the sequential linear rank statistic is key to deriving asymptotic properties of the test statistic. Some extensive simulations have been carried out to compare the operating characteristics of the method under different scenarios, and for a comparison with the usual log-rank test with fixed sample design. The methodology is also illustrated with a colon cancer clinical trial.

KEY WORDS: Clinical trial; Group-sequential procedure; Self-design; Survival data with staggered entry; Weighted linear rank statistic.

1. INTRODUCTION

In many clinical trials, it is often of interest to compare the duration of time to an event between patient groups on two treatment arms. Most such clinical trials are designed with a fixed number of patients and/or number of events of interest. Various authors have presented methods for the estimation of sample size or power under a usual fixed sample design (e.g., Lachin 1981; Rubinstein, Gail, and Santner 1981; Schoenfeld 1983; Kim and Tsiatis 1990). However, because of uncertainty in the initial estimates of patient variation, treatment effect, and compliance, it is desirable to adjust the sample size and/or the duration of the trial for more flexibility and adaptability. It is thus appealing to use the information from the current study at interim stages, updating the initial assumptions and adjusting the sample size if necessary, to ensure adequate power to detect a clinically meaningful treatment difference while maintaining the type I error rate.

Sample size reestimation, or self-designing methods, have been investigated over the last decade by many authors (e.g., Colton and McPherson 1976; Gould 1992; Shih 1992; Bauer and Kohne 1994; Proschan and Hunsberger 1995; Fisher 1998; Wassmer 1998; Cui, Hung, and Wang 1999; Shen and Fisher 1999; and Liu and Chi 2001). However, all of the methods have been developed for continuous or binary outcomes with immediately observed responses, and none is directly useful for survival response with staggered entry.

With staggered entry and long-term follow-up, partial but increasing information becomes available for the study participants at successive monitoring times. The information is contained not only in the sample size, but also in the number of events observed during the study. As acknowledged by Oakes (2001), development of sequential methods is difficult with time-to-event outcomes, because the martingale structure that underlies most techniques for survival analysis relates to the internal time scale for subjects and does not apply to the monitoring time scale. Over the past 20 years, various investigators have developed some basic theories for sequential designs and tests for censored survival data (e.g., Tsiatis 1982; Slud and Wei 1982; Tsiatis, Rosner, and Tritchler 1985; Lan and Lachin 1990; Lin 1991; Lin, Yao, and Ying 1999). Additionally, Andersen (1987) has shown how stochastic curtailed testing can be used to decide whether to continue a clinical trial using conditional power calculations based on the current data. However, the method itself cannot be used to extend the sample size without inflating the type I error rate.

Here we implement and illustrate a design and inference method of updating sample size using a nonparametric weighted linear rank test, based on the idea of Fisher (1998) and Shen and Fisher (1999). Specifically, we are concerned with incorporating patients entering into the trial at different times (staggered entry). Because of the uncertainty of the failure rate in either study group at the planning stage of the trial, we would like to use cumulative information at each intermediate stage of the current study to adjust the sample size sequentially. This procedure will ensure that the objective of the study can be accomplished with adequate power.

The article is organized as follows. In Section 2 we introduce a weighted rank test statistic that can update the sample size based on the observed data. We derive the asymptotic distribution for the proposed test under sequential monitoring. Its implementation relies on a weight function, which we propose in Section 3. The weight function at each stage is determined by the prior data, but not by current and future data. The design also allows the trial to be terminated early for futility, and we discuss two futility boundaries in Section 4. To achieve the specified power, the design may be modified by different means. In Section 5 we present a few strategies for adjusting a design based on the duration of the accrual, the duration of additional follow-up, or the accrual rate, so that a necessary number of events is observed during the entire study. We investigate the performance of the method by a series of simulation studies in Section 6, and illustrate an application of the proposed design method in Section 7 using a colon cancer clinical trial. We conclude with a discussion in Section 8.

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2. ASYMPTOTIC DISTRIBUTION OF THE STATISTIC

We consider the setting where patient entry is staggered, random, and independent of the survival and censoring times (other than administrative censoring at the end of the trial). After an initial accrual period, t_a, patients are followed for an additional period, t_f. The total study duration is $t_a + t_f$. We assume that the two treatment groups under study are entered sequentially and allocated symmetrically.

During the accrual phase, n individuals enter. For $i = 1, 2, \ldots, n$, let Y_i denote the real time of entry, let C_i denote the time from entry to censoring (e.g., loss to follow-up), let V_i denote the time from entry to the failure and let Z_i be the indicator variable for the treatment group for the ith patient. Then $\{(Y_i, V_i, C_i, Z_i),\ i = 1, \ldots, n\}$ are regarded as n independent and identically distributed (iid) vectors. In addition, we assume that given covariate Z, the random variables Y, V, and C are conditionally independent.

Suppose that a one-sided test will be performed with a significance level of α and a power of $(1-\beta)$ to detect a log-hazard ratio of $\delta > 0$. The null hypothesis of no treatment effect in terms of the log-hazard ratio θ is

$$H_0 : \theta = 0 \quad \text{versus} \quad H_1 : \theta > 0.$$

When data are examined at time t, we observe the time to failure or censoring, $X_i(t)$, and the failure indicator $\Delta_i(t)$, where

$$X_i(t) = \max\{\min(V_i, t - Y_i, C_i), 0\},$$

and $\Delta_i(t) = 1$ if $V_i \le \min(t - Y_i, C_i)$ and 0 otherwise. When reviewed at time t, the following variables can be represented as n iid random vectors $\{X_i(t), \Delta_i(t), Z_i\}$ for $i = 1, \ldots, n$.

Assume that we review the trial design after observing B_j events during the jth observation period $(t_{j-1}, t_j]$, where $t_0 = 0$ and $j = 1, 2, \ldots$. The number of events in the jth block $\{B_j,\ j = 1, 2, \ldots\}$ is specified before the trial starts, whereas the review time $t_j (j \ge 1)$ is a function of B_j. The study is carried out so that after observing every B_j failures, a linear rank test statistic, $U(t_j)$, based on all cumulated data up to time t_j, is computed and compared with a futility boundary. Specifically,

$$\begin{aligned} U(t) &= \sum_{i=1}^{n} \Delta_i(t)\left\{Z_i - \frac{S^{(1)}(t, X_i(t))}{S^{(0)}(t, X_i(t))}\right\} \\ &= \sum_{i=1}^{n} \int_0^t \left\{Z_i - \frac{S^{(1)}(t, x)}{S^{(0)}(t, x)}\right\} N_i(t, dx), \end{aligned}$$

where $N_i(t, x) = I(X_i(t) \le x, \Delta_i(t) = 1)$ and

$$S^{(r)}(t, x) = \frac{1}{n}\sum_{i=1}^{n} I\{X_i(t) \ge x\} Z_i^{\otimes r} \qquad (r = 0, 1).$$

If $U(t_j)$ is below the futility boundary, then the trial is terminated to accept H_0. Such a futility boundary precludes unnecessary extension if the experimental treatment is ineffective or inferior to the standard treatment. Otherwise, a weight function is estimated based on the data cumulated before the current step and is assigned to the current block of data. The procedure is iterated until the weight function is used up at certain step, J, when $\sum_{j=1}^{J-1} w_j^2 < 1$ and $\sum_{j=1}^{J} w_j^2 \ge 1$. When all of the weight function is spent at step J, the trial is terminated and H_0 tested for rejection or acceptance. In summary, the trial is terminated either when the weight is used up or when the futility boundary is crossed. A significance test statistic is computed and compared with the rejection boundary only at the last step. J is not fixed in advance, but rather is determined by observed data.

To review the cumulated data sequentially and propose the final test statistic, we need to derive the joint distribution of the statistic $U(t)$ over multiple (calendar) times. Assume that the trial is assessed after observing every B_j events between t_{j-1} and t_j, where $(t_1 < t_2 < \cdots < t_J)$. We approximate statistic, $U(t_j)$, by a sum of iid random variables and derive the asymptotic distribution of the final test statistic with the application of the multivariate central limit theorem. Under the null hypothesis, $U(t)$ can be approximated by $\widetilde{U}(t)$,

$$\widetilde{U}(t) = \sum_{i=1}^{n} \int_0^t \left\{Z_i - \frac{s^{(1)}(t, x)}{s^{(0)}(t, x)}\right\} M_i(t, dx) \equiv \sum_{i=1}^{n} q_i(t),$$

where

$$\begin{aligned} s^{(r)}(t, x) &= E\{S^{(r)}(t, x)\} \qquad (r = 0, 1), \\ M_i(t, x) &= N_i(t, x) - \int_0^x I\{X_i(t) \ge u\}\lambda(u)\, du, \\ q_i(t) &= \int_0^t \left\{Z_i - \frac{s^{(1)}(t, x)}{s^{(0)}(t, x)}\right\} M_i(t, dx), \end{aligned}$$

and $\lambda(\cdot)$ is the common hazard function of the failure time (see Lin 1991 for a similar derivation). For fixed t, $M_i(t, x)$ is a square-integrable martingale with respect to the filtration

$$\mathcal{F}_i(t) = \{\mathcal{F}_i(t, x); 0 \le x \le t\},$$

where $\mathcal{F}_i(t, x)$ is the sub-σ field generated by

$$\begin{aligned} \sigma\{&I(Y_i \le t), Z_i I(Y_i \le t), Y_i I(Y_i \le t), \\ &I[V_i \le \min(u, t - Y_i, C_i)], \\ &I[C_i \le \min(u, t - Y_i, V_i)] : 0 \le u \le x\}. \end{aligned}$$

Application of martingale stochastic integral theorems shows that $n^{-1/2}\widetilde{U}(t) \to N(0, \sigma^2(t))$, where $\sigma^2(t) = E\{q_1^2(t)\}$ and $E(q_1(t)) = 0$, and $n^{-\frac{1}{2}}\{U(t) - \widetilde{U}(t)\}$ converges in probability to 0. It then follows from the multivariate central limit theorem and the Cramer–Wold device that

$$n^{-\frac{1}{2}}\{U(t_i), U(t_j)\}, \qquad \text{for } i \ne j,$$

converges to a bivariate normal distribution with mean 0 and covariance

$$\sigma(t_i, t_j) = E\{q_1(t_i) q_1(t_j)\} = E\left\{q_1^2(t_i)\right\} \qquad \text{for } t_i < t_j$$

(see Tsiatis 1981 or Lin 1991 for a similar proof). As verified by Tsiatis (1982), a structure of independent increments in t arises for log-rank statistics when the intervals are defined by fixed calendar times. With staggered entry, the asymptotic joint distribution of the linear rank statistic, evaluated at times defined by observing prespecified numbers of events, has the same structure of independent increments (Gail, DeMets, and Slud 1982). Thus, for $i \ne j$,

$$\operatorname{cov}\{U(t_j) - U(t_{j-1}), U(t_i) - U(t_{i-1})\} \approx 0.$$

Following the foregoing properties, under the null hypothesis, the distribution of $n^{-\frac{1}{2}}\{U(t_j) - U(t_{j-1})\}$ converges to a normal distribution with mean 0 and variance $\sigma_j^2 - \sigma_{j-1}^2$, where $\sigma_j^2 \equiv \sigma^2(t_j) = \sigma(t_j, t_j)$.

It is natural to estimate the covariance elements $\sigma(t_i, t_j)$ by replacing the corresponding quantities with their empirical estimators. In particular, $\hat{\sigma}_j^2 = n^{-1}\sum_l Q_l(t_j)Q_l(t_j)$, where

$$Q_l(t) = \Delta_l(t)\left\{Z_l - \frac{S^{(1)}(t, X_l(t))}{S^{(0)}(t, X_l(t))}\right\} - \sum_{j=1}^{n} \frac{I\{X_j(t) \le X_l(t)\}\Delta_j(t)}{nS^{(0)}(t, X_j(t))}\left\{Z_l - \frac{S^{(1)}(t, X_j(t))}{S^{(0)}(t, X_j(t))}\right\}.$$

Note that the variance estimator for the log-rank test of Fleming and Harrington (1991, p. 261) is asymptotically equivalent to the foregoing estimator. The consistency of $\hat{\sigma}_j^2$ can be established by the properties of empirical distribution functions. Therefore, under H_0, the standardized information cumulated during (t_{j-1}, t_j) can be expressed as

$$S(t_j) \equiv n^{-1/2}\{U(t_j) - U(t_{j-1})\}\big/\left\{\hat{\sigma}_j^2 - \hat{\sigma}_{j-1}^2\right\}^{1/2}, \quad j = 1, 2, \ldots, \tag{1}$$

which is asymptotically normal, with mean 0 and $\mathrm{var}(S(t_j)) = 1$ by Slutsky's theorem. Here we assume that $U(t_0) = 0$ and $\hat{\sigma}_0 = 0$. Furthermore, for any given i, j $(i \ne j)$, using an approach similar to that of Tsiatis (1982), we can show that the random vector $\{S(t_i), S(t_j)\}$ converges in distribution to a standard bivariate normal distribution by a direct application of the Cramer–Wold device. Thus the normalized score statistics $S(t_i)$ and $S(t_j)$ are asymptotically independent for any $i \ne j$.

Applying Theorem 2.1 of Shen and Fisher (1999), let T_J denote the final test statistic, which has the form

$$T_J = \sum_{j=1}^{J} w_j S(t_j),$$

where $w_j = w_j(S(t_1), \ldots, S(t_{j-1}))$ is a nonnegative weight function consisting of data up to the $(j-1)$th step. Note that J is a random integer determined by prior data. The selection of the weight function is such that with probability 1, there exists a positive finite number J such that $\sum_{j=1}^{J} w_j^2 = 1$. Thus T_J is asymptotically normally distributed with mean 0 and variance 1 under the null hypothesis. Based on this property and the strategy that the null hypothesis is not being rejected in the intermediate stage (i.e, when $\sum_{i=1}^{j} w_i^2 < 1$), the type I error rate is not inflated, but it may be deflated due to futility stopping. The important structural feature of T_J is that $S(t_j)$ can be approximated by a sum of martingales with asymptotic independent increments to which the central limit theorem and the theorem of Shen and Fisher (1999) apply. The next critical step is estimation of the weight function w_j at each stage, which is discussed in the following section.

3. DERIVATION OF WEIGHT FUNCTION

In an analysis of time-to-event data, the number of events carries primary information. Thus derivation of the weight function is based on the number of events, rather than on the actual sample size. An important role of the weight function in the test statistic is to extend or contract the number of events required over time based on cumulated information. To accomplish this, we would like to spend more weight for the next block to complete the trial sooner when prior data suggest a strong indication that the specified power to reject H_0 will be achieved, or vice versa. Thus it is natural to propose a weight function to be an inverse function of the additional events required to achieve the specified power. The additional events needed to achieve the specified power can be derived from the conditional power equation for the log-rank test.

Let $d(t_j)$ denote the expected number of events at time t_j. The weight function, w_j, for $j > 1$ is estimated iteratively using observed data prior to the jth look. For $j = 1$, we assign a fixed fraction (e.g., 1/2) to w_1 at the first look. Given $\mathcal{F}_i(t_{j-1})$, $i = 1, 2, \ldots$, let $t_j^\star$ be the calendar time when we should have sufficient number of events to achieve the specified power and to terminate the trial. Note that $t_j^\star$ is different from t_j in general. To ensure the specified power $1 - \beta$ given the accumulated data up to step $(j-1)$, we solve the additional number of events needed, $d(t_j^\star) - d(t_{j-1})$, from the conditional power equation,

$$P\left(\sum_{i=1}^{j-1} w_i S_i + w_j S_j^\star > z_\alpha \mid \theta = \hat{\theta}_{j-1}\right) = 1 - \beta,$$

where $S_j^\star$ is the standardized increment of the log-rank statistic between t_{j-1} and $t_j^\star$. The left side of the foregoing equation can be rewritten as

$$P\left(S_j^\star - \sqrt{n}\hat{\theta}_{j-1}\sqrt{\sigma_j^2 - \sigma_{j-1}^2} > \frac{z_\alpha - \sum_{i=1}^{j-1} w_i S_i}{w_j} - \sqrt{n}\hat{\theta}_{j-1}\sqrt{\sigma_j^2 - \sigma_{j-1}^2} \mid \theta = \hat{\theta}_{j-1}\right).$$

Note that $S_j^\star$ is asymptotically independent of $S_1, \ldots, S_{j-1}$, and $\sigma^2(t_i)$ can be approximated by $\sigma_z^2 d(t_i)/n$, where σ_z^2 is the variance of treatment indicator Z (Tsiatis et al. 1985). Assuming that the jth step is the last step of the trial, we have that $w_j = \sqrt{1 - \sum_{i=1}^{j-1} w_i^2}$. By solving the equation

$$\frac{z_\alpha - \sum_{i=1}^{j-1} w_i S_i}{\sqrt{(1 - \sum_{i=1}^{j-1} w_i^2)}} - \sqrt{d(t_j^\star) - d(t_{j-1})}\hat{\theta}_{j-1}\sigma_z = z_{1-\beta},$$

we obtain $d(t_j^\star) - d(t_{j-1})$ as a function depending on data observed prior to the jth step,

$$d(t_j^\star) - d(t_{j-1}) = \left\{\frac{z_\alpha - \sum_{i=1}^{j-1} w_i S_i}{\sqrt{1 - \sum_{i=1}^{j-1} w_i^2}} + z_\beta\right\}^2 \Bigg/ \hat{\sigma}_z^2\hat{\theta}_{j-1}^2,$$

where z_α and z_β denote the upper α- and β-standard normal percentiles, and $\hat{\sigma}_z$ is the empirical estimator of σ_z. In the foregoing equation, if the estimated additional number of events is smaller than 1, then we exit the trial after observing the next block of data and using all of the remaining weight; otherwise, we propose a weight function for the jth block data as follows. With an estimated additional number of events required between t_{j-1} and $t_j^\star$, it is natural to equally allocate the weight, $(1 - \sum_{i=1}^{j-1} w_i^2)$ that is left after $(j-1)$ steps, to subjects in the jth block. In particular, the weight function at the jth step is proposed as an inverse function of $d(t_j^\star) - d(t_{j-1})$,

$$w_j = \sqrt{\frac{B_j(1 - \sum_{i=1}^{j-1} w_i^2)}{d(t_j^\star) - d(t_{j-1})}}, \qquad \text{for } j > 1,$$

where B_j is a prespecified number of events for the jth block. It is clear that w_j depends only on data prior to the jth step. Note that the additional number of events, $d(t_j^\star) - d(t_{j-1})$, is used simply as a bridge to estimate the weight for the jth step data, not for deciding the number of events needed for the jth block.

4. FUTILITY STOPPING BOUNDARY

For both ethical and economic reasons, it is practical to monitor a trial and stop it if the experimental treatment is ineffective or inferior to the standard one. Moreover, such an early stopping boundary can be useful in constraining an unnecessary increase of sample size in the self-designing setting under the null hypothesis and preserving the type I error rate. Using ideas similar to those of Shen and Fisher (1999), we use the following two boundaries for futility stopping: Wald-type constant likelihood boundary and the lower boundary of the confidence interval specified at significance level α_0,

$$f_1(t_j) = n^{1/2}\sigma_j \left\{ \frac{\delta_j}{2} + \log\left(\frac{\beta}{1-\alpha}\right) \Big/ \delta_j \right\} \tag{2}$$

and

$$f_2(t_j) = n^{1/2}\sigma_j \left(\delta_j + z_{1-\alpha_0}\right), \tag{3}$$

where $\delta_j = \sqrt{n}\delta\sigma_j$ is a noncentrality parameter. When $U(t_j) \le f_1(t_j)$ (or $U(t_j) \le f_2(t_j)$), the trial is terminated because of concerns that the new therapy may be ineffective or inferior in prolonging the survival time of patients. The early stopping boundary, f_1, is derived from DeMets and Ware's (1980) Wald-type constant likelihood boundary for a normal distribution with unit variance and a noncentrality parameter. In the current context, $n^{-1/2}U(t_j)/\sigma_j$ is asymptotically normal with unit variance and noncentrality parameter δ_j. Because such a boundary depends on the alternative hypothesis δ, it would reduce the length of the study under the H_0 and still have good properties under the alternative hypothesis. The second boundary, f_2, which is derived from the upper boundary of the $(1-\alpha_0)\%$ confidence interval of θ, is more intuitive than f_1. Specifically, we stop the trial and accept H_0 if the upper boundary of the confidence interval of θ with level $(1-\alpha_0)$ is less than δ. Here the confidence interval of θ is estimated using data observed up to time t_j. It is clear that there is no concern for an inflated type I error rate caused by this type of interim analyses. Such early termination even leads to a deflation of the type I error rate from the specified nominal level, as we will show in the simulation.

5. STRATEGIES FOR ADJUSTING A DESIGN

Theoretically, the self-designing method would allow the sample size, or the number of events for censored survival data, to increase to whatever value indicated. In practice, clinical trials have limitations in resources that can be reflected by a limit on the number of patients, the duration of the trial, or the accrual rate.

Typically in a clinical trial, patients enter sequentially and are followed until they fail or the study is terminated. Based on the accumulated information, the researchers decide to continue the trial or to terminate it due to futility or when the weight is used up. When it is decided to continue a trial, researchers can extend the follow-up duration after accrual, t_f, the accrual period, t_a, or increase the accrual rate, C. Factors that determine the number of failures are the hazard rates for both treatment arms, the accrual rate, the accrual time, and the followup duration after accrual. In particular, we have proposed three strategies for adjusting the study design under the self-designing framework.

Assume that an expected treatment effect δ is tested by a log-rank test based on the partial likelihood (Schoenfeld 1983). Then for equal randomization with fixed sample design, the total number of events required to achieve $(1-\beta)$ power at the significance level α is given by the expression

$$d = \frac{4(z_\alpha + z_\beta)^2}{\delta^2}, \tag{4}$$

where δ is the log-hazard ratio.

Although the power is determined by the number of events in survival analysis, it is still important to estimate the number of patients required. Assume that patient accrual is uniform during $(0, t_a)$ with an accrual rate C, and that survival distributions for the two treatment arms are exponential with rates λ_0 and λ_1. The expected number of failures by time t in each arm is

$$d_{\lambda_i}(t) = \frac{C}{2}\left\{ t - \frac{1 - e^{-\lambda_i t}}{\lambda_i} \right\} I\{t \le t_a\} + \frac{C}{2}\left\{ t_a - \frac{e^{-\lambda_i t}}{\lambda_0}(e^{\lambda_i t_a} - 1) \right\} I\{t > t_a\}, \qquad i = 0, 1.$$

By letting $d(t) = d_{\lambda_0}(t) + d_{\lambda_1}(t)$, we can easily solve for the accrual rate C. Given an accrual duration t_a, the required sample size is equal to Ct_a.

We now describe three strategies for updating a design. We first consider the setting with a fixed accrual rate and accrual duration (and hence fixed sample size), while letting the follow-up duration vary. By extending the follow-up time, t_f, we may observe extra failures when necessary. It is intuitive that with low hazard rates for the study population, this type of adjustment may be attractive, because fewer failures will be observed during an early stage of the trial, but more events should be observed in a later period with extended t_f.

Second, we assume that the accrual rate and the follow-up duration after accrual are fixed. Under the self-designing setting, when it is necessary to increase the number of events, we will extend the accrual duration. This strategy of design can be more effective when hazard rates for the study population are relatively high, but one can only make inference about the short-term treatment effect if more patients are recruited but followed for relatively short time.

Finally, we consider the setting with a fixed maximum study duration. If it is feasible to adjust the accrual rate, then we may then fix both accrual duration and follow-up duration, t_a and t_f, but increase C when necessary. This strategy may be less practical, because it is often difficult to substantially change the accrual rate in the practice of clinical trials.

6. SIMULATION

A typical clinical trial with censored survival data as primary outcomes has four key quantities in its design: (1) number of failures, (2) accrual rate, (3) accrual duration, and (4) follow-up duration. In the simulations, we explore the three strategies discussed in Section 5. For simplicity, we assume that censoring occurs only administratively at each look and at the end of the trial. The empirical type I error rates and power estimates are based on simulating 1,000 independent trials for various combinations of accrual rate, accrual period, follow-up duration, and hazard rates.

Assume that patients enter the trial with a Poisson process at a fixed rate per unit time. Survival times for patients are generated as exponentially distributed random variables. Patients are randomly assigned to one of two treatment groups with a probability of .5. Power of 90% with a one-sided type I error rate of .025 is specified. As found by Shen and Fisher (1999), the weight function can be sensitive to the variations caused by the data when the block size is small, especially for the first block. Thus we use a relatively large block size at the first look to obtain a reasonable estimate for the variance of the statistic. In particular, we use one-half of the number of events required in the fixed sample design as B_1, and a constant block size of event of 4, 5, or 6 for B_i, $i > 1$. In each simulation, a usual log-rank test is computed based on the calculated sample size from the fixed sample design at the prespecified terminating time. The performance of the log-rank test from the fixed sample design is compared with the proposed procedures in terms of type I error rate, power, average number of events, and average study duration by simulation. Note that in the simulation, we used n_j instead of n in the estimation of variance of $U(t_j)$, $S(t_j)$ in (1), and futility boundary (2) and (3), where n_j is the number of independent subjects enrolled into the study up to the calendar time t_j.

We first consider the situation with fixed accrual rate and accrual period, but with follow-up duration varying from 2 to 8 years. Assume a fixed accrual duration of 2 years and an accrual rate of 126 per year. With an expected log-relative risk under the alternative hypothesis, $\delta = \log(2) = .693$ with $\lambda_0 = .2$, $\lambda_1 = .1$, the required sample size is 252 and the number of events is 88 for the fixed sample design with a 2-year follow-up after accrual. If the underlying distributions are correctly specified, then the foregoing design will have a power of 90% to detect a log-relative risk of log(2) with a significance level of .025 during a total study duration of 4 years. The usual log-rank test statistic is calculated at the end of 4 years for 252 accrued subjects.

For the proposed method, a log-rank score statistic, $U(t_j)$, is calculated after observing $\sum_{i=1}^{j} B_i$ failures at time t_j, by resorting the follow-up times of all patients who had entered the trial up to t_j. At each look, subjects who entered the trial at that time but have not failed are censored. If $U(t_j)$ is below the futility boundary, f_1, then the trial is terminated to accept H_0 at t_j. Otherwise, a weight function is calculated based on the data observed prior to the jth step. If the cumulative summation of the weight up to the jth step is less than 1, then the study will continue; otherwise, the study is finished at the jth step. Some preliminary simulations showed that the Wald-type constant likelihood boundary, f_1, was very conservative; therefore, we adjust the boundary slightly as $\sqrt{n}\sigma_j\{\delta_j/3 + \log(\frac{\beta}{1-\alpha})/\delta_j\}$ in simulations.

The empirical sizes given in Table 1 show that the proposed test statistics under the null hypothesis are fairly conservative,

Table 1. Empirical Size/Power and Average Number of Events for a Design With One-Sided $\alpha = .025$, $1 - \beta = .9$, Futility Boundary f_1, and Alternative Hypothesis $H_1: \delta = \log(2) = .693$ (with $\lambda_0 = .2$ and $\lambda_1 = .1$)

	Proposed method			*Fixed-sample design*
	B = 4	*B = 5*	*B = 6*	
Under the null $H_0: \theta = 0$				
Empirical size	.010	.012	.014	.020
Average number of events	72.29	73.40	74.26	113.08
Average study duration	2.86	2.90	2.93	4.00
Underlying log-hazard ratio $\theta = .693$				
Empirical power	.904	.885	.903	.897
Average number of events	90.78	89.18	91.22	88.80
Average study duration	4.10	4.12	4.07	4.00
Underlying log-hazard ratio $\theta = .598$				
Empirical power	.858	.850	.860	.805
Average number of events	109.76	111.48	110.86	91.24
Average study duration	5.38	5.48	5.43	4.00
Underlying log-hazard ratio $\theta = .553$				
Empirical power	.815	.816	.814	.729
Average number of events	117.84	115.86	118.17	92.88
Average study duration	5.76	5.64	5.74	4.00
Underlying log-hazard ratio $\theta = .511$				
Empirical power	.756	.751	.749	.675
Average number of events	120.23	120.66	119.44	94.29
Average study duration	5.85	5.85	5.73	4.00

NOTE: The accrual duration and accrual rate are fixed, and t_f can be varied from 2 to 8 years.

Table 2. Empirical Size/Power and Average Number of Events for a Design With One-Sided $\alpha = .025$, $1-\beta = .9$, Futility Boundary f_1, and Alternative Hypothesis $H_1: \delta = \log(2.5) = .916$ (with $\lambda_0 = .5$, $\lambda_1 = .2$)

	Proposed method			*Fixed-sample design*
	$B=4$	$B=5$	$B=6$	
Under the null $H_0: \theta = 0$				
Empirical size	.012	.014	.015	.025
Average number of events	44.24	44.90	45.07	66.09
Average study duration	2.39	2.41	2.41	4.00
Underlying log-hazard ratio $\theta = .916$				
Empirical power	.916	.920	.922	.899
Average number of events	58.13	57.60	57.42	52.56
Average study duration	3.38	3.37	3.35	4.00
Underlying log-hazard ratio $\theta = .777$				
Empirical power	.889	.878	.894	.801
Average number of events	69.23	68.33	68.72	54.15
Average study duration	3.75	3.70	3.71	4.00
Underlying log-hazard ratio $\theta = .693$				
Empirical power	.786	.803	.808	.709
Average number of events	74.93	74.92	73.20	55.38
Average study duration	3.89	3.89	3.82	4.00
Underlying log-hazard ratio $\theta = .616$				
Empirical power	.706	.717	.704	.598
Average number of events	76.10	77.82	77.24	56.48
Average study duration	3.90	3.92	3.90	4.00

NOTE: The accrual rate and t_f are fixed, t_a can be varied from 2 to 4 years.

with an average number of events only 65% of that from the fixed sample design. When the alternative hypothesis is correctly specified, the proposed procedure achieves power similar to that from the fixed sample design with a slightly greater number of events. When the true underlying alternatives, θ, are overestimated by δ and are .598, .553, and .511, corresponding to $\lambda_0 = .2$, $\lambda_1 = .11$, .115, and .12, the test statistics based on the fixed sample design result in a substantial loss of power. In contrast, the proposed test statistics that allow an increase in the number of events by extending t_f are effective to achieve a larger power. In some scenarios, the proposed method gains 12% more power than the fixed-sample design.

We then consider the situation with fixed accrual rate and t_f, but the accrual duration, t_a, may be changed from 2 years to 4 years. As discussed in Section 5, this type of strategy for adjusting the number of events may work more effectively when hazard rates are relatively high. With a fixed hazard rate in the placebo group, $\lambda_0 = .5$, the hazard rates in the treatment groups are .2, .23, .25, and .27 for four scenarios, corresponding to log-hazard ratios of .916, .777, .693, and .616. The results are summarized in Table 2. Here $\delta = .916$ (with $\lambda_1 = .2$) is assumed to be the alternative hypothesis. In the case where the hazard ratio is considerably overestimated compared with the actual data, the power yielded from the fixed sample design is about 60%, whereas the new method can achieve as much as 72% power. Notably, the gain in power can be close to 20%.

To compare the current strategy with strategy 1, we also evaluated the performance by extending t_a given the relatively low hazard rates as in Table 1. Results in Table 3 show that extending t_a seems to be quite effective for both of the hazard rates that we studied here. Moreover, a slightly greater gain in power is achieved by extending t_a rather than t_f. As expected, the average number of events used also increases with this strategy. In addition, we considered extending t_f for the case with higher hazard rates in the study population, as in Table 2; only a small gain in power is achieved by just extending t_f compared with the fixed sample design. We do not present the simulation results here, for space considerations. The explanation is that most events occur during an early stage of the study, so an extension of the follow-up will not yield many additional events.

Finally, we present simulation results for the fixed maximum duration design with a possibly varying accrual rate in Table 4. Under the same specifications as those in Table 2 ($\lambda_0 = .5$ vs. $\lambda_1 = .2, \ldots, .27$), the proposed method is effective in increasing power compared with the fixed sample design, but is slightly less efficient compared with the strategy of extending t_a. However, the difference between the two strategies is not substantial in terms of overall performance. With lower hazard rates ($\lambda_0 = .2$ vs. $\lambda_1 = .1, \ldots, .12$) such as in Table 1, we also assess the strategy of varying the accrual rate by fixing the total study duration. The results (not presented due to similarity) show that the performance is similar to that reported in Table 1. As suggested by the associate editor, we have also performed additional simulations with smaller treatment effects. In particular, with fixed $t_a = 2$ and $t_f = 3$, the expected log-relative risk is $\delta = \log(1.67) = .511$ with $\lambda_0 = .5$. The true underlying hazard rate in the treatment group (λ_1) varies from .3 to .36. The results, summarized in Table 5, show that the improvement in power is similar to that seen in the settings with large treatment effects.

Among all three strategies, block size may not greatly affect the performance of the proposed methods. However, the constant log-likelihood futility boundary outperforms the lower boundary of the confidence interval, especially when the magnitude of overestimating the underlying treatment difference is large (the results for using f_2 are not shown). Under the null hypothesis, the proposed procedures across all three strategies use less resources in terms of the average number of events observed and average maximum study duration.

Table 3. Empirical Size/Power and Average Number of Events for a Design With One-Sided $\alpha = .025$, $1-\beta = .9$, Futility Boundary f_1, and Alternative Hypothesis $H_1: \delta = \log(2) = .693$ (with $\lambda_0 = .2$, $\lambda_1 = .1$)

	Proposed method			Fixed-sample design
	$B = 4$	$B = 5$	$B = 6$	
Under the null $H_0: \theta = 0$				
Empirical size	.024	.021	.025	.030
Average number of events	76.40	76.81	77.45	113.20
Average study duration	2.58	2.59	2.61	4.00
Underlying log-hazard ratio $\theta = .693$				
Empirical power	.930	.934	.928	.908
Average number of events	103.31	104.03	101.42	88.32
Average study duration	3.58	3.60	3.55	4.00
Underlying log-hazard ratio $\theta = .598$				
Empirical power	.885	.881	.890	.816
Average number of events	122.54	120.98	119.59	91.19
Average study duration	3.92	3.88	3.85	4.00
Underlying log-hazard ratio $\theta = .553$				
Empirical power	.850	.865	.857	.774
Average number of events	124.80	123.66	120.45	93.50
Average study duration	3.91	3.89	3.82	4.00
Underlying log-hazard ratio $\theta = .511$				
Empirical power	.801	.786	.795	.666
Average number of events	137.82	140.07	137.59	94.13
Average study duration	4.13	4.18	4.13	4.00

NOTE: The t_f and accrual rate are fixed, the accrual duration t_a can be varied from 2 to 4 years.

7. EXAMPLE

To illustrate the method, we use a completed multicenter randomized clinical trial as an example. This trial was designed as a fixed-sample study. Our purpose was to show what would have happened had the trial been designed to allow sample size adjustment in doing inference under the self-designing framework.

Patients with resected stages B and C colorectal carcinoma were randomized to placebo, adjuvant levamisole (lev), or adjuvant levamisole plus fluorouracil (lev + 5-FU) for 1 year after surgery. One of the primary goals was to evaluate the efficacy of levamisole, either alone or in combination with 5-FU, to reduce recurrence of the disease. Some detailed analyses have been given by Moertel et al. (1990) and Laurie et al. (1989). For the purposes of illustration here, we restrict attention to the efficacy comparison between the placebo and the lev + 5-FU arm. The variable of interest is time to recurrence of disease or death, whichever occurs first.

Among the total of 619 patients entered to the trial between the years 1984 and 1987, 189 out of 315 patients in the placebo arm and 134 out of 304 in the lev + 5-FU arm relapsed or died, with a median of 7.5 years of follow-up. Using the usual log-rank test in the fixed sample design, the observed differ-

Table 4. Empirical Size/Power and Average Number of Events for a Design With One-Sided $\alpha = .025$, $1-\beta = .9$, Futility Boundary f_1, and Alternative Hypothesis $H_1: \delta = \log(2.5) = .916$ (with $\lambda_0 = .5$, $\lambda_1 = .2$)

	Proposed method			Fixed-sample design
	$B = 4$	$B = 5$	$B = 6$	
Under the null $H_0: \theta = 0$				
Empirical size	.011	.013	.015	.027
Average number of events	44.96	45.49	45.54	66.01
Average study duration	1.63	1.64	1.64	4.00
Underlying log-hazard ratio $\theta = .916$				
Empirical power	.928	.927	.930	.903
Average number of events	54.93	54.95	55.08	52.17
Average study duration	2.27	2.27	2.27	4.00
Underlying log-hazard ratio $\theta = .777$				
Empirical power	.852	.847	.850	.801
Average number of events	64.07	64.11	64.84	54.37
Average study duration	2.47	2.47	2.48	4.00
Underlying log-hazard ratio $\theta = .693$				
Empirical power	.764	.763	.775	.708
Average number of events	68.32	67.93	69.32	55.76
Average study duration	2.56	2.55	2.59	4.00
Underlying log-hazard ratio $\theta = .616$				
Empirical power	.692	.689	.694	.619
Average number of events	72.43	73.04	72.32	56.80
Average study duration	2.64	2.65	2.63	4.00

NOTE: The t_a and t_f are fixed, and the accrual rate can be varied.

Table 5. Empirical Size/Power and Average Number of Events for a Design With One-Sided $\alpha = .025$, $1-\beta = .9$, Futility Boundary f_1, and Alternative Hypothesis $H_1: \delta = \log(1.67) = .511$ (with $\lambda_0 = .5$, $\lambda_1 = .3$)

	Proposed method			Fixed-sample design
	$B=4$	$B=5$	$B=6$	
Under the null $H_0: \theta = 0$				
Empirical size	.015	.017	.011	.020
Average number of events	133.0	133.7	135.5	182.2
Average study duration	1.87	1.87	1.89	5.00
Underlying log-hazard ratio $\theta = .510$				
Empirical power	.920	.916	.912	.895
Average number of events	179.0	180.2	178.3	164.5
Average study duration	2.70	2.70	2.67	5.00
Underlying log-hazard ratio $\theta = .416$				
Empirical power	.835	.827	.835	.747
Average number of events	209.5	209.3	207.5	168.3
Average study duration	3.02	3.02	3.00	5.00
Underlying log-hazard ratio $\theta = .386$				
Empirical power	.764	.768	.761	.691
Average number of events	220.1	215.8	220.4	169.2
Average study duration	3.15	3.08	3.13	5.00
Underlying log-hazard ratio $\theta = .329$				
Empirical power	.640	.639	.637	.554
Average number of events	234.1	235.3	233.8	171.3
Average study duration	3.29	3.30	3.27	5.00

NOTE: The t_a and t_f are fixed, and the accrual rate can be varied.

ence of disease-free survival between the placebo arm and the lev + 5-FU arm is statistically significant with a one-sided p value of .0001.

Suppose that the investigators would like to detect a 50% decrease in the hazard rate in the treatment group compared with the placebo group, in which patients have a hazard rate of .23, assuming an exponential distribution. These assumptions are reasonably close to the protocol with a fixed-sample design (Laurie et al. 1989) and the observed data. Given an accrual duration of 4 years and an additional 5 years of follow-up, a sample size of 619 would have about 80% power to detect a 50% decrease of the hazard rate in recurrence and/or death. If the self-designing procedure with flexible accrual rate and a block size of 5 at each look after the first step were applied, then the trial would have been terminated with the null hypothesis being rejected after 47 events had occurred, that is, on completion of the third look. In this case, a total of 314 patients had been entered to the study, and the total study duration was 2.5 years. Specifically, the trial would be stopped earlier than with the fixed-sample design and would claim the efficacy of levamisole together with 5-FU, compared with the placebo. The inference based on the self-designing trial results in a test statistic of 3.1 with a p value of .001.

It is interesting to note that using O'Brien and Fleming's sequential design with a maximum of four looks, the trial would be terminated early at the second interim analysis to reject the null hypothesis when 100 events were observed (O'Brien and Fleming 1979). Conversely, using Pocock's sequential procedure with a maximum of four looks, the trial would be stopped to reject the null hypothesis after the first interim analysis after 59 failures were observed (Pocock 1977).

8. CONCLUDING REMARKS

In this article we have derived and illustrated a flexible design and inference procedure with updating of the sample size for censored survival data with a staggered entry. The proposed test statistic is a nonparametric weighted linear rank statistic. One virtue of the approach is that the sample size and/or the number of events of the proposed weighted test statistic is determined sequentially based on cumulated information to ensure adequate power. With such a flexible method, it is important for ethical and economic reasons to not extend a trial if the experimental treatment is ineffective or inferior to the standard one. Therefore, an early termination scheme is implemented in the procedure to prevent the unneeded extension of the trial, while maintaining the type I error rate. As one reviewer pointed out, if the experimental treatment is grossly inferior to the standard one, then the first planned interim analysis for futility may not be early enough. Hence if investigators have a strong belief that such a possibility exists, we recommend they impose an additional monitoring boundary for futility before the first planned interim analysis. This type of stopping rule would not inflate the overall type I error rate.

There are two keys that allow us to generalize the self-design method of Shen and Fisher (1999) to censored survival data with staggered entry. First, in the sequential setting, the log-rank-type statistics can be considered in the same way as partial sums of independent normal observations asymptotically. In particular, our techniques involve an approximation of the score process by a sum of iid martingales, and these statistics have uncorrelated increments in chronological time. Moreover, in the derivation of weight function, it is crucial that we have used the approximation that the variance of the score statistic is proportional to the expected number of failures.

The test statistic derived in this article is based on a commonly used log-rank test statistic; however, it should easily extend to any other types of weighted rank statistics, and even to nonrank statistics, as long as the statistics have uncorrelated increments. Furthermore, even when trials involve multiple endpoints, the same inference strategy can be generalized to the linear rank statistics with respect to the marginal distributions of the multiple endpoints (Lin 1991), if such linear rank statistics have independent increments.

We have proposed three strategies for adjusting the number of events in the framework of the self-designing clinical trial, and have compared them in certain scenarios. Based on our limited simulations, the strategy of continuing a trial by extending the accrual period, t_a, whereas fixing t_f and the accrual rate seems to work more effectively than the other two strategies for a study population with the hazard rates that we have considered. It is clear that there are many other possible combinations of strategies for adjusting a study design. Some trial and error in the selection of a strategy can be helpful. In practice, it is reasonable to perform a small-scale simulation study to select the most economical and efficient plan based on the prior information for the hazard rates of the study population and practical constraints in terms of time and expense.

The proposed approach is promising in restoring power by extending a current study, and thereby shortening the drug development process, for the "almost significant" trial while preserving the type I error rate. This procedure should be more cost-effective than the alternative of carrying out another large study based on observed information. However, the current procedure may not be optimal nor completely efficient; therefore, it would be interesting to investigate more efficient weight functions and strategies. In a regulatory setting, blinded sample size reestimation methods are preferred (Gould 1992; Shih 1992) over unblinded methods. A further investigation may study the flexible sample size reestimation method in this framework while blinded to treatment assignment. Moreover, it would be interesting to combine some features of the commonly used group-sequential designs with the sample size reestimation strategies. Finally, it is worth generalizing the method by using blocks defined by calendar time rather than the number of events for planned interim analyses.

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